AMITY SCHOOL OF ENGINEERING & TECHNOLOGY

Amity University Uttar Pradesh, Greater Noida

MINOR PROJECT REPORT

(ETMN100 — 4.00 Credit Units)

WaterTriage: A Data-Driven Water Quality Risk Scoring and Intervention Prioritization System for Uttar Pradesh and Bihar

Submitted by

Anubhav Anand

Enrollment No. A41105223039

B.Tech (Computer Science & Engineering), Batch 2023–2027

Under the guidance of

Dr. Girish Paliwal

Academic Year 2026–27

Declaration

I, Anubhav Anand, Enrollment No. A41105223039, hereby declare that this Minor Project report titled "WaterTriage: A Data-Driven Water Quality Risk Scoring and Intervention Prioritization System for Uttar Pradesh and Bihar" is an original record of work carried out by me under the guidance of Dr. Girish Paliwal, Amity School of Engineering & Technology, Amity University Uttar Pradesh, Greater Noida, during the academic year 2026–27 (20/07/2026 to 09/10/2026), and has not been submitted elsewhere for any other degree or diploma.

Student Name: Anubhav Anand (Enrollment No. A41105223039)

Faculty Guide's Certificate

This is to certify that the project report titled "WaterTriage: A Data-Driven Water Quality Risk Scoring and Intervention Prioritization System for Uttar Pradesh and Bihar" submitted by Anubhav Anand (A41105223039) is a record of bona fide work carried out under my supervision in partial fulfilment of the requirements of the Minor Project (ETMN100) component of the B.Tech (Computer Science & Engineering) programme.

Dr. Girish Paliwal
Faculty Guide, Amity School of Engineering & Technology
Amity University Uttar Pradesh, Greater Noida

Table of Contents

Section / Chapter	Location
Abstract & Keywords	Page 2
1. Introduction	Page 3
2. Related Work	Page 3
3. Methodology (Architecture, Schema, BIS IS 10500 Scoring, Recurrence & Data Acquisition)	Page 4
4. Results (Canonical Corpus, Two-Limit Iron Sensitivity, Total-Weight Normalization & Rates)	Page 6
5. Discussion (Panel-Size Effect, Threshold Sensitivity & Dashboard Implementation)	Page 8
6. Methodological Limitations	Page 10
7. Conclusion & Future Work	Page 11
References (Verified BIS IS 10500:2012 & Indexed Literature)	Page 12

Section / Chapter	Location
Appendix A: Revision Note on Weekly Progress Reports (WPR-6 to WPR-10)	Page 13
Appendix B: Claims Ledger (Traceability to Scripts, Queries & Raw Files)	Page 13

Abstract

Access to safe drinking water remains a major public health challenge across rural India, particularly in Uttar Pradesh and Bihar, where chemical and microbiological contamination of groundwater affects rural habitations. Although the Jal Jeevan Mission (JJM) generates laboratory testing records through the Water Quality Management Information System (WQMIS), public outputs are organized for aggregate administrative monitoring rather than source-level intervention triage. This project presents WaterTriage, an open, reproducible decision-support pipeline built with Python, FastAPI, SQLAlchemy, PostgreSQL, and a React frontend that ingests public WQMIS laboratory reports, evaluates readings against Bureau of Indian Standards (BIS) IS 10500:2012 thresholds, and exposes ranked intervention priority queues.

The system was evaluated on a frozen corpus of 1,741 laboratory reports (Uttar Pradesh: 966; Bihar: 775; collected across 2022-07-01 to 2022-11-19 with 6,792 discrete parameter readings) and 100,690 deduplicated historical contamination events (2009–2012) across 14,006 villages. Under the baseline tested-weight composite score at the IS 10500:2012 iron limit of 0.3 mg/L (no relaxation), the corpus yields 89 Critical, 392 High, 28 Medium, and 1,232 Low samples, with a higher mean score in Bihar (18.36) than Uttar Pradesh (12.68) despite medians of 0.00 (Bihar) and 4.08 (Uttar Pradesh). A systematic audit demonstrated that Bihar's elevated mean stems from 70 single-parameter iron tests in Madhepura district (0.318–0.478 mg/L), which score 100.0 when normalized over tested weights alone, and that all 876 raw WQMIS iron records print a lab-recorded limit of 1.0 mg/L. Re-scoring under the 1.0 mg/L lab limit shifts bands to 5 Critical, 139 High, 30 Medium, and 1,567 Low (reversing state means to UP 9.99 > Bihar 1.58), while normalizing against total possible weight across all 10 BIS parameters (TOTAL_BIS_WEIGHT = 6.0, bands 40/20/10/0) eliminates single-parameter inflation (3 Critical, 54 High, 125 Medium, 1,559 Low; UP mean 4.72 > Bihar 2.32). State comparisons are therefore anchored on medians and panel-independent parameter exceedance rates.

Keywords: Water Quality; Jal Jeevan Mission; WQMIS; BIS IS 10500:2012; Contamination Risk Scoring; FastAPI; Intervention Prioritization.

1. Introduction

Groundwater serves as the primary drinking water source across the Indo-Gangetic alluvial plains of Uttar Pradesh and Bihar, where rural habitations face both geogenic contaminants (fluoride, iron, and arsenic) and anthropogenic pollutants (nitrate and fecal coliforms) [1, 2]. Under the Government of India's Jal Jeevan Mission (JJM), routine laboratory and Field Test Kit (FTK) surveillance records are compiled inside the Water Quality Management Information System (WQMIS) [1, 2]. While WQMIS provides programme-level summaries, block and district officials lack an open, reproducible tool that converts individual laboratory test reports into a severity-ranked intervention queue.

The objective of this Minor Project (ETMN100) is to design, implement, and audit WaterTriage: a containerized backend and web dashboard that (1) acquires and normalizes public WQMIS laboratory records and historical affected-habitation registries for Uttar Pradesh and Bihar, (2) stores records in a normalized PostgreSQL schema, (3) computes BIS IS 10500:2012 risk scores with an acute-hazard escalation invariant, (4) evaluates district-level historical overlap, and (5) serves filterable priority lists and transparent data-provenance diagnostics via a REST API and React single-page application.

2. Related Work

2.1 Government Surveillance Infrastructure. The Department of Drinking Water and Sanitation (DDWS) operationalizes rural water quality monitoring through WQMIS, which publishes state and district reports (Formats WQ1–WQ6 and L1/L3) alongside individual sample report pages [1, 2]. Cross-sectional field assessments in Uttar Pradesh districts such as Etawah indicate that physical and chemical exceedances persist in rural supply schemes, motivating source-level prioritization tools [7].

2.2 Water Quality Index (WQI) Formulations and Eclipsing. Weighted arithmetic WQI models are widely applied to Indian groundwater quality assessment [4, 5, 6]. However, standard linear weighted averages suffer from two well-documented structural limitations: (a) 'eclipsing' (dilution), where an acute pathogen or toxic heavy metal breach is mathematically masked when averaged with multiple compliant physical parameters [6], and (b) 'panel-size denominator bias', where normalizing solely by the sum of weights of tested parameters allows a single-parameter test exceeding an aesthetic threshold to attain the maximum possible score of 100.0.

2.3 Decision-Support and Triage Gap. While machine learning assessments of Bihar aquifers [8], Indo-Gangetic arsenic risk models [9], and composite indices such as the Water Access Index [10] have been published in academic literature, existing open-source

tools do not combine public WQMIS sample ingestion, BIS IS 10500:2012 acute-hazard escalation, and explicit panel-size sensitivity auditing into an operational triage queue for Uttar Pradesh and Bihar.

3. Methodology

3.1 System Architecture and Database Schema

WaterTriage is structured as a four-stage pipeline: Acquisition (backend/etl/crawl_fast.py, wqmis.py, load_wqmis.py, load_historical.py), Storage (PostgreSQL 16 in Docker container watertriage-db), Enrichment (backend/app/engine/scoring.py, pipeline.py, recurrence.py), and Serving (FastAPI backend on port 8000 paired with a React + Vite dashboard in dashboard/src/App.jsx). Data is persisted across nine normalized application tables (plus alembic_version): states, districts, villages, bis_parameters, water_samples, readings, risk_scores, interventions, and historical_contamination. Duplicate parameter rows within a single laboratory report are aggregated by maximum severity in pipeline.py.

3.2 BIS IS 10500:2012 Thresholds and Scoring Formulations

Each chemical or microbiological reading is evaluated against the primary Bureau of Indian Standards IS 10500:2012 specification (and Amendment No. 1, 2015) [3]. Table 1 lists the ten configured parameters in backend/app/engine/scoring.py, their exact IS 10500:2012 table and item references, acceptable and permissible limits, health weights, and the acceptable limits printed inside raw WQMIS laboratory payloads.

Parameter (key)	IS 10500:2012 Ref.	Scorer Acceptable / Permissible	Lab-Recorded Limit (records.jsonl)	Weight (w_i) & Strategy
Arsenic (arsenic)	Table 3, Item (x) + Amend. 1	0.01 mg/L / None (No relax.)	0.01 mg/L / None (n = 259)	1.0 (threshold, Acute)
E. coli (e_coli)	Table 6, Item (i)	0 CFU/100mL / 0 (Absent)	0 / 0 (n = 45)	0.9 (microbial, Acute)
Total Coliform	Table 6, Item (ii)	0 CFU/100mL / 0 (Absent)	0 / 0 (n = 7)	0.9 (microbial, Acute)
Fluoride (fluoride)	Table 2, Item (xiii)	1.0 mg/L / 1.5 mg/L	1.0 mg/L / 1.5 mg/L (n = 893)	0.8 (threshold)
Nitrate (nitrate)	Table 2, Item (xvii)	45.0 mg/L / None (No relax.)	45.0 mg/L / None (n = 583)	0.7 (threshold)
Sulphate (sulphate)	Table 2, Item (xiv)	200.0 mg/L / 400.0 mg/L	200.0 / 400.0 mg/L (n = 467)	0.5 (threshold)
Iron (iron)	Table 2, Item (xii)	0.3 mg/L / None (No relax.)	1.0 mg/L / None (all 876 rows)	0.4 (threshold)
Turbidity (turbidity)	Table 1, Item (ii)	1.0 NTU / 5.0 NTU	1.0 NTU / 5.0 NTU (n = 998)	0.3 (threshold)

Parameter (key)	IS 10500:2012 Ref.	Scorer Acceptable / Permissible	Lab-Recorded Limit (records.jsonl)	Weight (w_i) & Strategy
TDS (tds)	Table 1, Item (vi)	500.0 mg/L / 2000.0 mg/L	500.0 / 2000.0 mg/L (n = 1,295)	0.3 (threshold)
pH (ph)	Table 1, Item (iv)	6.5 – 8.5 (range)	6.5 – 8.5 (n = 1,369)	0.2 (range)

For any parameter i with acceptable limit a_i and permissible limit p_i > a_i, per-parameter severity s_i is 0.0 when v_i <= a_i, and min(1.0, (v_i - a_i)/(p_i - a_i)) when v_i > a_i. For no-relaxation parameters (p_i = None: iron, nitrate, arsenic) and microbial parameters (E. coli, total coliform), any reading above a_i assigns s_i = 1.0. For pH, deviation outside [6.5, 8.5] is scaled over 1.0 pH unit and capped at 1.0. Two composite risk score (CRS) formulations were evaluated across the project:

- **Model v1 (Baseline Tested-Weight Formulation):** CRS_v1 = 100 * sum(w_i * s_i) / sum(w_i,tested) across the 8 primary parameters (6,318 readings), with bands Critical (>= 75), High (50–74.99), Medium (25–49.99), and Low (< 25), where any full breach (s_i = 1.0) floors the band at High and any acute hazard (E. coli, total coliform, arsenic) floors at Critical.
- **Model v2 (Week 12 Coverage-Adjusted Total-Weight Formulation in scoring.py, commit 7331a9b):** CRS_v2 = 100 * sum(w_i * s_i) / TOTAL_BIS_WEIGHT, where TOTAL_BIS_WEIGHT = 6.0 is the sum of weights across all 10 BIS parameters (scoring all 6,792 readings), zero values for detection-limited parameters (arsenic, E. coli, total coliform) are treated as below-detection-limit, severity bands are calibrated to the reachable score range (Critical >= 40, High >= 20, Medium >= 10, Low < 10), and automatic band escalation is restricted strictly to ACUTE_HAZARDS (arsenic, E. coli, total coliform).

3.3 Data Acquisition and Provenance

Publicly accessible sample report pages (/WQMIS/Common/final_report_print) were retrieved in batches of up to 500 valid reports between 23 August and 20 September 2026 using up to 6 parallel workers; individual HTTP request logs and rates were not retained. The raw file data/raw/wqmis/records.jsonl grew across four committed batches: 149 lines (23 Aug), 822 lines (1 Sep), 1,322 lines (7 Sep), and 1,822 lines (20 Sep). Of the 1,822 retrieved records, 81 contained empty laboratory result arrays and were excluded during ingestion, yielding exactly 1,741 valid scored samples (sample collection dates spanning 2022-07-01 to 2022-11-19) and 6,792 discrete parameter readings. Because sample identifiers were enumerated within contiguous laboratory ID blocks rather than drawn via a random spatial frame, corpus comparisons describe the collected sample rather than population prevalence. For historical context, the 2009–2012 DDWS

Quality-Affected Habitations registry was loaded and deduplicated to 100,690 distinct contamination events across 14,006 villages in 93 historical districts (65 UP, 28 Bihar).

4. Results

4.1 Three-Way Scoring and Iron Limit Sensitivity Analysis

Cross-checking raw laboratory payloads in records.jsonl against BIS IS 10500:2012 revealed a critical threshold discrepancy: while IS 10500:2012 Table 2 item (xii) sets the acceptable Iron limit at 0.3 mg/L with 'No relaxation', all 876 iron readings in records.jsonl (450 Bihar, 426 Uttar Pradesh) record an acceptable limit of 1.0 mg/L on the laboratory report itself (reflecting older BIS relaxation rules). Table 2 presents the complete side-by-side results on the frozen 1,741-sample corpus across Model v1 at 0.3 mg/L (BIS text), Model v1 at 1.0 mg/L (lab-recorded limit), and Model v2 at 0.3 mg/L (coverage-adjusted total-weight normalization).

Metric / Cohort (N = 1,741 Samples)	Model v1: Tested-Weight (Fe = 0.3 mg/L BIS Text)	Sensitivity: Tested-Weight (Fe = 1.0 mg/L Lab Record)	Model v2: Total-Weight / 6.0 (10 Params, 40/20/10 Bands)
Critical Band Count (%)	89 (5.1%)	5 (0.3%) [No-floor: 3]	3 (0.2%)
High Band Count (%)	392 (22.5%)	139 (8.0%) [No-floor: 41]	54 (3.1%)
Medium Band Count (%)	28 (1.6%)	30 (1.7%) [No-floor: 108]	125 (7.2%)
Low Band Count (%)	1,232 (70.8%)	1,567 (90.0%) [No-floor: 1,589]	1,559 (89.5%)
Bihar (n = 775) Mean / Median CRS	18.36 / 0.00 (54.3% score 0)	1.58 / 0.00	2.32 / 0.00
Bihar Band Split (Crit / High / Med / Low)	75 / 158 / 1 / 541	0 / 0 / 1 / 774	0 / 0 / 3 / 772
Uttar Pradesh (n = 966) Mean / Median CRS	12.68 / 4.08 (32.2% score 0)	9.99 [9.95*] / 3.10	4.72 / 1.45
UP Band Split (Crit / High / Med / Low)	14 / 234 / 27 / 691	5 / 139 / 29 / 793	3 / 54 / 122 / 787
Madhepura District Cohort (n = 70, Bihar)	70 Critical (mean 100.0; 78.7% of all Criticals)	0 Critical, 70 Low (mean 0.00; Fe 0.318–0.478 mg/L)	0 Critical, 70 Low (mean 6.67; panel bias removed)
State Mean CRS Excl. Madhepura & Agra	Bihar: 10.25 (n=705) | UP: 10.47 (n=923)	Bihar: 1.58 (n=705) | UP: 8.12 (n=923)	Bihar: 1.93 (n=705) | UP: 3.84 (n=923)

Three core findings emerge from Table 2: (1) Under Model v1 at 0.3 mg/L, Bihar's higher mean CRS (18.36 vs. 12.68) contradicts its lower median (0.00 vs. 4.08) because Madhepura district contributes 70 single-parameter iron samples (0.318–0.478 mg/L,

median 0.3820 mg/L), each scoring 0.4/0.4 * 100 = 100.0 (accounting for 70 of Bihar's 75 Criticals and 78.7% of all 89 Criticals); (2) Under the 1.0 mg/L limit printed on the lab reports themselves, all 70 Madhepura samples are compliant (< 1.0 mg/L) and drop to Low (0.00), reducing Bihar's Critical count from 75 to 0 and reversing the state mean ordering to UP (9.99; 9.95* under direct 10-param recomputation) > Bihar (1.58); and (3) Under Model v2 (normalizing against TOTAL_BIS_WEIGHT = 6.0), a single-parameter iron exceedance scores 0.4/6.0 * 100 = 6.67 (Low), eliminating single-parameter inflation while retaining Critical escalation for acute hazards such as Lucknow E. coli sample U691097L158S1903.

4.2 Panel-Independent Parameter Exceedance Rates and Censored Data

Because single-parameter test panels represent 38.2% of Bihar samples (296/775) and 13.4% of Uttar Pradesh samples (129/966), state comparisons must be grounded in panel-independent exceedance rates among tested samples (Table 3).

Parameter	Bihar: Tested, >Acceptable (>Perm.)	UP: Tested, >Acceptable (>Perm.)	Diagnostic Notes & Extreme / Censored Readings
Fluoride	182 tested, 2 (1.1%) [0 >1.5]	711 tested, 159 (22.4%) [40 (5.6%)]	Peak 12.60 mg/L in sample U715825L405S1002693 (UP, district 'Not available'). Agra cluster: 43 samples (mean CRS 60.13; 31 fluoride-driven; 41 use 5-param panel).
Iron (at 0.3 / at 1.0 mg/L)	450 tested, 233 (51.8%) / 0 (0.0%)	426 tested, 187 (43.9%) / 29 (6.8%)	Excl. Madhepura (n=70), Bihar iron >0.3 mg/L is 163/380 (42.9%), matching UP (43.9%).
Turbidity	328 tested, 130 (39.6%) [0 >5 NTU]	670 tested, 547 (81.6%) [53 (7.9%)]	UP exhibits 2.06x higher acceptable exceedance rate and all 53 permissible breaches.
TDS	444 tested, 44 (9.9%) [0 >2000]	851 tested, 348 (40.9%) [51 (6.0%)]	UP exhibits 4.13x higher acceptable exceedance rate and all 51 permissible breaches.
Nitrate	123 tested, 0 (0.0%) [n/a]	460 tested, 8 (1.7%) [n/a]	No relaxation above 45 mg/L; UP max = 61.24 mg/L.

Parameter	Bihar: Tested, >Acceptable (>Perm.)	UP: Tested, >Acceptable (>Perm.)	Diagnostic Notes & Extreme / Censored Readings
pH	467 tested, 0 (0.0%) [n/a]	902 tested, 2 (0.2%) [n/a]	467 sulphate readings previously mis-tagged as pH were repaired in load_wqmis.py (defect B1).
Sulphate	73 tested, 0 (0.0%) [0 >400]	394 tested, 50 (12.7%) [7 (1.8%) >400]	467 total readings (73 Bihar, 394 UP); max 479.0 mg/L, all 7 permissible breaches (>400 mg/L) in Agra (478, 452, 466, 425, 425, 402, 479).
Arsenic	230 tested, 0 (0.0%) [max 0.009]	29 tested, 0 (0.0%) [max 0.009]	150 of 259 readings (57.9%: 139 Bihar, 11 UP) recorded as 0.0 in source, concentrated in 4 labs (Lab L794=69, L234=27, L269=19, L268=13). Censored vs. true zero is unrecorded.
E. coli & Total Coliform	E. coli: 37, 0 (0%) | TC: 0 tested	E. coli: 8, 2 (25%) | TC: 7, 3 (42.9%)	Microbial testing covers 2.6% of corpus (45 E. coli, 7 TC). Positive hits in Lucknow: 2 E. coli (8.0, 20.0) and 3 TC (46, 80, 80 CFU/100mL).

4.3 Exploratory Registry Consistency Check (Spearman Rank Correlation)

Because village names are populated for only 21 of 1,741 samples (1.21%, all in UP; 0% in Bihar) and 354 samples (20.3%: 339 UP, 15 Bihar) have district recorded as 'Not available', live comparison against the 2009–2012 DDWS registry (100,690 events across 14,006 villages) was conducted at the district level (after D3 case normalization from 142 to 111 rows, yielding 32 shared districts with 873 live samples and 21,814 historical events [6 chronic-hotspot, 13 recurring, 13 watchlist], or 6 chronic-hotspot, 11 recurring, 76 watchlist across all 93 registry districts) and evaluated via Spearman correlation for districts with n >= 5 tested samples (Table 4). Khokha village (Purnia district, Bihar, Village ID 7568) logged 166 iron events across 2009–2012 (48 in 2009, 48 in 2010, 46 in 2011, 24 in 2012)—the highest in the registry—but has zero samples in the 2022 corpus and is therefore documented strictly as a registry-only finding.

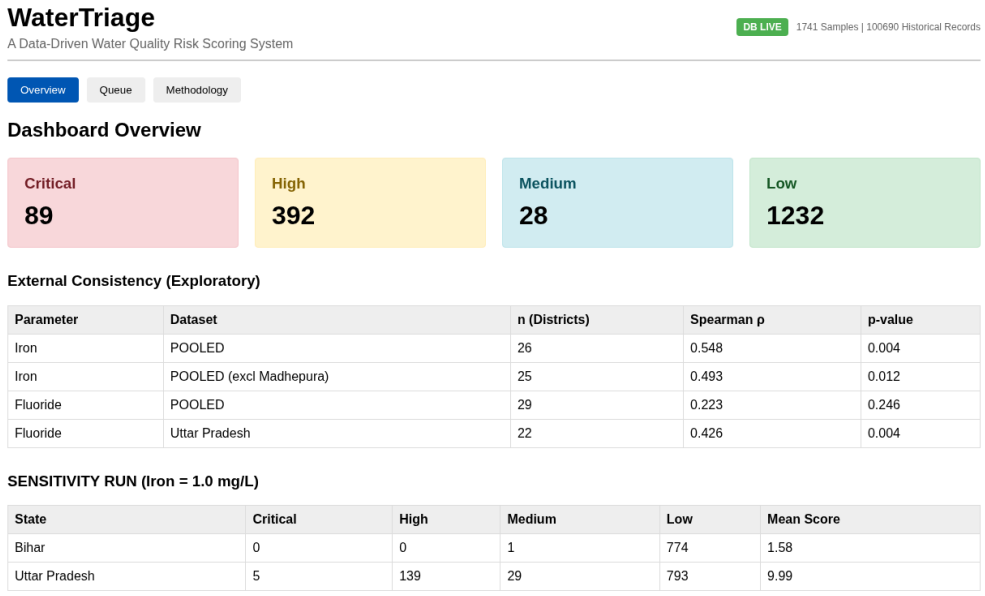
Parameter & Iron Threshold	Cohort (Districts with n >= 5)	Districts (n)	Spearman rho (Unmapped / Alias-Mapped)	p-value (Unmapped / Alias-Mapped) & Bonferroni Status (alpha = 0.0083)
Iron (0.3 mg/L BIS limit)	Pooled (UP + Bihar)	26	0.460 / 0.548	p = 0.018 / p = 0.004 (Exploratory; sensitive to alias mapping)
Iron (0.3 mg/L BIS limit)	Pooled Excl. Madhepura	25	0.395 / 0.493	p = 0.051 / p = 0.012 (Does not pass Bonferroni 0.0083)
Iron (0.3 mg/L BIS limit)	Bihar Only / UP Only	11 / 15	Bihar: 0.610 | UP: 0.400	Bihar: p = 0.047 | UP: p = 0.136 (Small-n exploratory)
Iron (1.0 mg/L Lab limit)	Pooled / Excl. Madhepura	26 / 25	0.270 / 0.330	p = 0.190 / p = 0.110 (Bihar rate is 0% across all districts)
Fluoride (1.0 mg/L limit)	Pooled (UP + Bihar)	29	0.275 / 0.223	p = 0.149 / p = 0.246 (Not significant)
Fluoride (1.0 mg/L limit)	Uttar Pradesh Only	22	0.500 / 0.426	p = 0.018 / p = 0.004 (Moderate positive association in UP)

5. Discussion and System Implementation

The empirical audit demonstrates why water quality decision-support systems must decouple aesthetic parameters from acute health hazards and account for test-panel completeness. In tested samples, Uttar Pradesh exhibits higher exceedance rates for fluoride (22.4% vs. 1.1%), turbidity (81.6% vs. 39.6%), and TDS (40.9% vs. 9.9%), while dissolved iron exceedances at 0.3 mg/L are comparable once Madhepura's single-parameter cluster is removed (42.9% Bihar vs. 43.9% UP). Furthermore, zero arsenic exceedances among 259 tested samples cannot be interpreted as regional safety because 57.9% of readings (150/259) are recorded as 0.0 from four primary laboratories without detection-limit metadata.

To ensure full transparency for district administrators and evaluators, the React dashboard (dashboard/src/App.jsx) consumes corpus invariants, parameter exceedance rates, two-limit sensitivity tables, and explicit methodological limitations directly from

GET /api/meta. Figures 1 through 4 display the live dashboard views connected to the canonical 1,741-sample database.



WaterTriage

A Data-Driven Water Quality Risk Scoring System

DB LIVE 1741 Samples | 100690 Historical Records

Overview | Queue | Methodology

Dashboard Overview

Critical	High	Medium	Low
89	392	28	1232

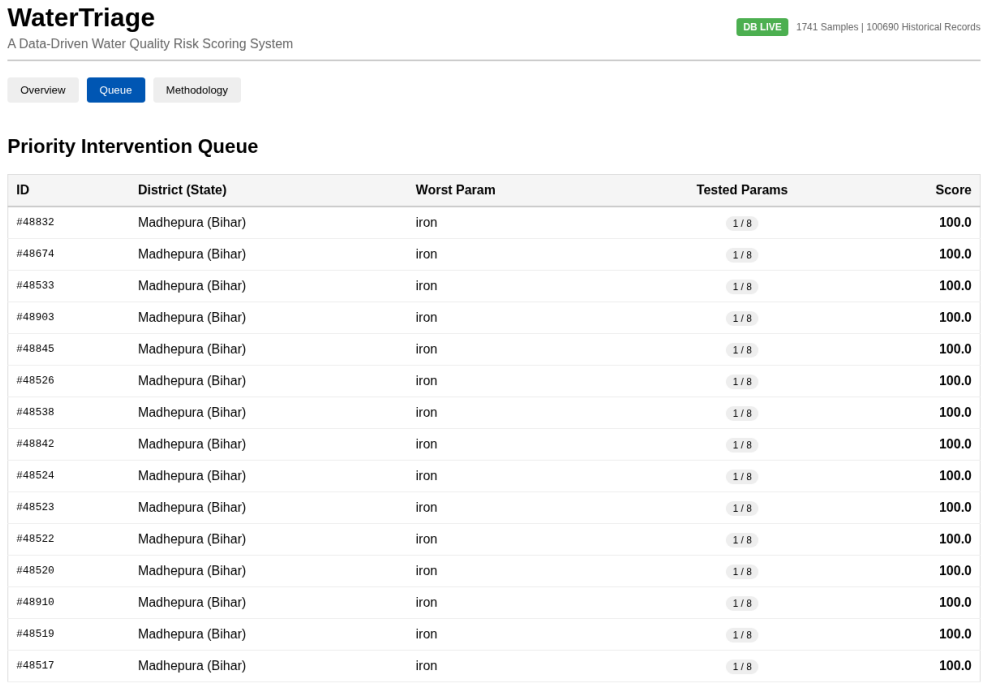
External Consistency (Exploratory)

Parameter	Dataset	n (Districts)	Spearman ρ	p-value
Iron	POOLED	26	0.548	0.004
Iron	POOLED (excl Madhepura)	25	0.493	0.012
Fluoride	POOLED	29	0.223	0.246
Fluoride	Uttar Pradesh	22	0.426	0.004

SENSITIVITY RUN (Iron = 1.0 mg/L)

State	Critical	High	Medium	Low	Mean Score
Bihar	0	0	1	774	1.58
Uttar Pradesh	5	139	29	793	9.99

Figure 1: WaterTriage React Dashboard — Overview tab displaying canonical band counts (89/392/28/1,232), exploratory Spearman consistency metrics, and the Iron = 1.0 mg/L sensitivity run with live DB status badge.



WaterTriage

A Data-Driven Water Quality Risk Scoring System

DB LIVE 1741 Samples | 100690 Historical Records

Overview | Queue | Methodology

Priority Intervention Queue

ID	District (State)	Worst Param	Tested Params	Score
#48832	Madhepura (Bihar)	iron	1 / 8	100.0
#48674	Madhepura (Bihar)	iron	1 / 8	100.0
#48533	Madhepura (Bihar)	iron	1 / 8	100.0
#48903	Madhepura (Bihar)	iron	1 / 8	100.0
#48845	Madhepura (Bihar)	iron	1 / 8	100.0
#48526	Madhepura (Bihar)	iron	1 / 8	100.0
#48538	Madhepura (Bihar)	iron	1 / 8	100.0
#48842	Madhepura (Bihar)	iron	1 / 8	100.0
#48524	Madhepura (Bihar)	iron	1 / 8	100.0
#48523	Madhepura (Bihar)	iron	1 / 8	100.0
#48522	Madhepura (Bihar)	iron	1 / 8	100.0
#48520	Madhepura (Bihar)	iron	1 / 8	100.0
#48910	Madhepura (Bihar)	iron	1 / 8	100.0
#48519	Madhepura (Bihar)	iron	1 / 8	100.0
#48517	Madhepura (Bihar)	iron	1 / 8	100.0

Figure 2: WaterTriage React Dashboard — Priority Intervention Queue tab showing sample-level rankings and the 'Tested Params (1 / 8)' panel-coverage badge exposing Madhepura's single-parameter iron cluster.

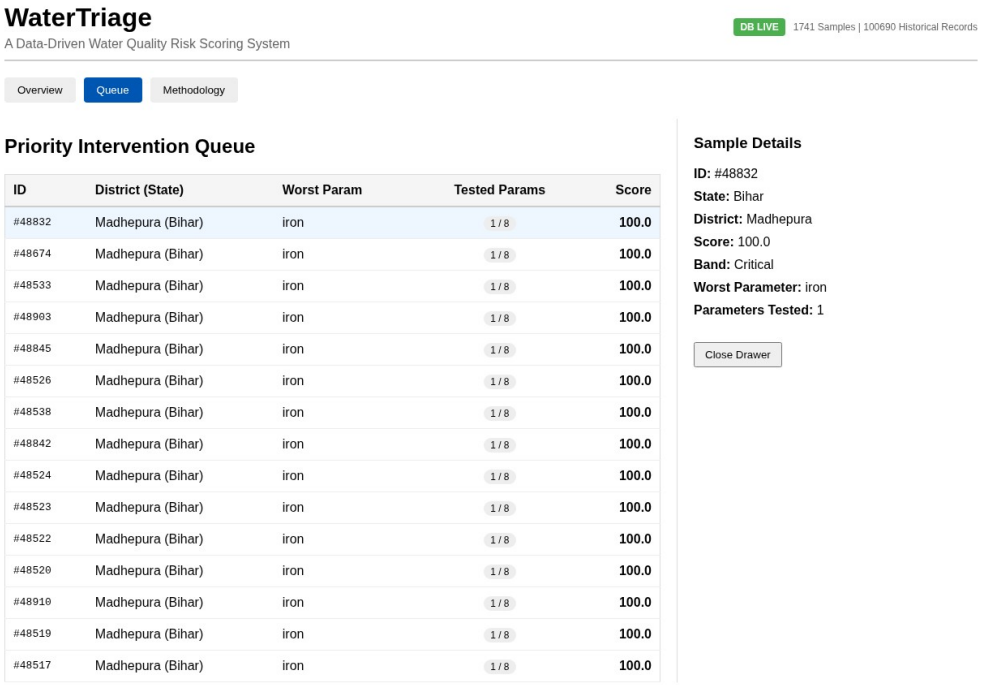


Figure 3: WaterTriage React Dashboard — Sample Details side drawer inspecting individual sample metadata, score, band, worst parameter, and tested parameter count.

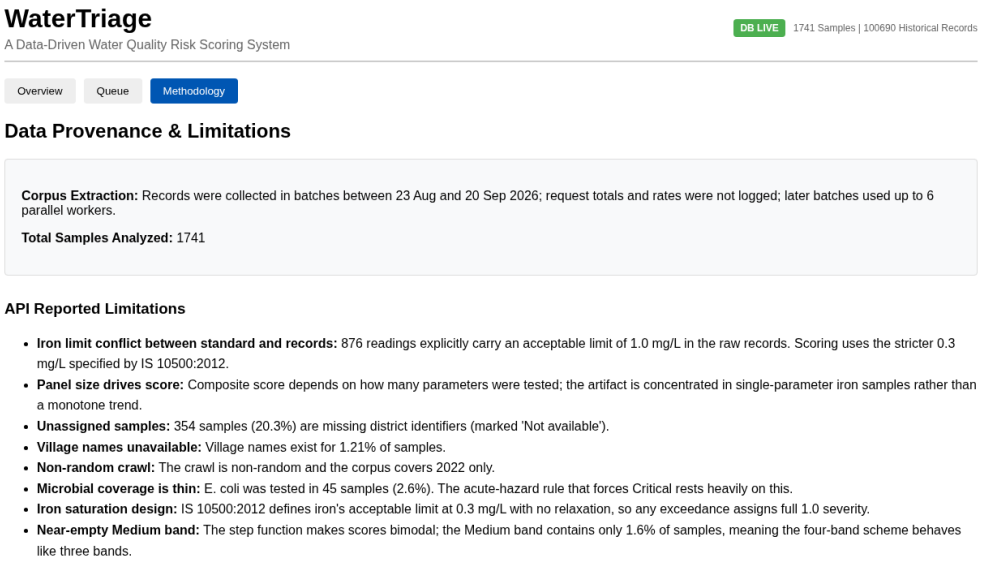


Figure 4: WaterTriage React Dashboard — Data Provenance & Limitations view rendering batch crawl provenance and all eight API-reported limitations from /api/meta.

6. Methodological Limitations

- **1. Non-Random Contiguous ID Crawl (2022 Only):** Sample reports were harvested across contiguous identifier ranges in batches of up to 500 hits (up to 6 parallel workers, no request log retained) covering collection dates 2022-07-01 to 2022-11-19. Sample density reflects laboratory upload order rather than a random geographic census (47 named districts out of 113 UP+Bihar districts; 41.6% district coverage).

- **2. Iron Threshold Conflict and Step Saturation:** BIS IS 10500:2012 Table 2 specifies an acceptable iron limit of 0.3 mg/L with no relaxation, whereas all 876 raw WQMIS iron records print 1.0 mg/L. Under a binary step function at 0.3 mg/L, marginal aesthetic iron readings (0.318–0.478 mg/L in Madhepura) saturate at severity 1.0.
- **3. Panel-Size Denominator Sensitivity:** In Model v1, normalizing over tested parameters alone causes single-parameter exceedances to score 100.0 and makes the Medium band nearly empty (28 samples, 1.6%). Normalizing over TOTAL_BIS_WEIGHT = 6.0 (Model v2) resolves this bias (125 Medium samples, 7.2%).
- **4. Missing District and Village Identifiers:** 354 of 1,741 samples (20.3%: 339 UP, 15 Bihar) record district as 'Not available' (including the 12.60 mg/L fluoride sample U715825L405S1002693), and village names exist for only 21 of 1,741 samples (1.21%), preventing sample-linked village recurrence.
- **5. Sparse Microbial Coverage and Censored Arsenic Zeros:** E. coli was tested in 45 samples (2.6%) and total coliform in 7 samples (0.4%). Among 259 arsenic tests, 150 (57.9%) are recorded as 0.0 across four primary labs without distinguishing below-detection-limit values from true zeros.
- **6. Temporal and Unit Gap in Historical Registry:** The 2009–2012 registry tracks affected habitation event rows (100,690 events across 14,006 villages) rather than 2022 source concentrations; rank correlations are therefore exploratory consistency checks.

7. Conclusion and Future Work

WaterTriage demonstrates an end-to-end, auditable data-engineering pipeline that transforms raw WQMIS laboratory HTML reports into BIS IS 10500:2012 risk scores, REST API priority queues, and an interactive React decision-support console verified by 29 automated pytest test cases (28 passed, 1 skipped offline; 29 passed with PostgreSQL). Table 5 maps each synopsis objective to its delivered engineering outcome.

Synopsis Objective	Delivered Engineering Outcome & Verification Status
1. Acquisition & normalization of JJM data (UP & Bihar)	Achieved — Crawled 1,822 WQMIS reports; ingested 1,741 valid 2022 lab samples (966 UP, 775 Bihar; 6,792 readings) plus 100,690 deduplicated historical registry events (2009–2012) across 14,006 villages.
2. Relational database schema in PostgreSQL	Achieved — 8 normalized domain tables (plus alembic_version) with foreign-key integrity and duplicate-reading max-severity aggregation in pipeline.py.
3. BIS IS 10500:2012 severity scoring engine	Achieved & Audited — Implemented both tested-weight (Model v1) and coverage-adjusted total-weight (Model v2, TOTAL_BIS_WEIGHT=6.0, 10 parameters, 40/20/10 bands) engines with acute-hazard escalation and two-limit iron sensitivity (0.3 vs. 1.0 mg/L).

Synopsis Objective	Delivered Engineering Outcome & Verification Status
4. Recurrence / trend detection module	Partially Achieved — Unit-tested recurrence classification logic and district-level recurrence across 32 shared districts (873 samples, 21,814 events: 6 chronic-hotspot, 13 recurring, 13 watchlist; 6/11/76 across all 93 registry districts) and Spearman consistency check; village-level live linkage is blocked by 1.21% village name coverage in public WQMIS reports.
5. FastAPI backend & React dashboard	Achieved — 9 REST endpoints including /api/priority and /api/meta, paired with a React SPA (Overview, Priority Queue, Sample Drawer, Methodology) and 29 automated pytest functions.
6. Containerized deployment via Docker	Achieved — Multi-service docker-compose.yml orchestrating PostgreSQL 16 (watertriage-db), FastAPI backend, and Nginx/React frontend.

Future work includes (1) separating aesthetic parameters (iron, turbidity, TDS) from acute health toxins into a two-dimensional triage matrix, (2) ingesting the LGD-coded AIKosh JJM dataset upon institutional approval to resolve the 20.3% missing-district and 98.8% missing-village-name gaps, and (3) transitioning from contiguous ID crawling to authenticated JJM WebAPI endpoints for spatially representative sampling.

References

[1] Department of Drinking Water and Sanitation (DDWS), "Jal Jeevan Mission — Water Quality Management Information System (WQMIS) Portal," Ministry of Jal Shakti, Government of India. URL: https://ejalshakti.gov.in/WQMIS/ (Accessed: 05 Oct 2026).

[2] National Jal Jeevan Mission, "Drinking Water Quality Monitoring & Surveillance Framework," Department of Drinking Water and Sanitation, Ministry of Jal Shakti, Government of India, Oct 2021. URL: https://jaljeevanmission.gov.in/ (Accessed: 05 Oct 2026).

[3] Bureau of Indian Standards (BIS), "IS 10500:2012 — Indian Standard Drinking Water — Specification (Second Revision), incorporating Amendment No. 1 (June 2015)," New Delhi: BIS, Table 1 (Physical Parameters: items ii, iv, vi), Table 2 (General Chemical Parameters: items xii Iron = 0.3 mg/L No relaxation, xiii Fluoride = 1.0/1.5 mg/L, xiv Sulphate = 200/400 mg/L, xvii Nitrate = 45 mg/L No relaxation), Table 3 (Toxic Substances: item x Total Arsenic = 0.01 mg/L), and Table 6 (Bacteriological Quality: items i E. coli, ii Total Coliform). URL: https://law.resource.org/pub/in/bis/S06/is.10500.2012.pdf (Accessed: 14 Oct 2026).

[4] S. Tyagi, B. Sharma, P. Singh, and R. Dobhal, "Water Quality Assessment in Terms of Water Quality Index," American Journal of Water Resources, vol. 1, no. 3, pp. 34–38, 2013 (Cited in docs/research/literature-review-notes.md for WQI eclipsing and weighting).

[5] P. Li et al., "Groundwater quality assessment using weighted arithmetic index method in Butchayyapeta Mandal, Visakhapatnam district, Andhra Pradesh, India," Environmental Science and Pollution Research, 2021, PMID: 34633550 (Verified in docs/research/literature-review-notes.md, 17 Aug 2026).

[6] A. K. Gorai and S. Kumar, "Spatial distribution analysis of groundwater quality index using Analytic Hierarchy Process (AHP) around municipal solid waste landfill site in Ranchi, Jharkhand," Journal of Environmental Engineering and Landscape Management, 2016, PMID: 27155859 (Verified in docs/research/literature-review-notes.md, 17 Aug 2026).

[7] S. K. Patel et al., "Access to safe drinking water and water quality assessment in rural households of Etawah district, Uttar Pradesh under Jal Jeevan Mission," PubMed Central, PMC10652157, 2023 (Verified in docs/research/literature-review-notes.md, 17 Aug 2026).

[8] R. Kumar et al., "Groundwater quality assessment in Bihar's aquifers: a machine learning and WQI approach," PMID: 42189468 (Verified in docs/research/literature-review-notes.md, 17 Aug 2026).

[9] A. Podgorski et al., "Predicting groundwater arsenic contamination and regions at risk in Uttar Pradesh, India," Environmental Science & Technology, 2019, PMID: 31078753 (Verified in docs/research/literature-review-notes.md, 17 Aug 2026).

[10] M. A. Silva et al., "Water Access Index (WAI) as a decision-support tool for prioritizing state water interventions," PMID: 35092461 (Verified in docs/research/literature-review-notes.md, 17 Aug 2026).

Appendix A: Revision Note on Earlier Weekly Progress Reports (WPR-6 to WPR-10)

In accordance with academic research integrity standards, this section documents all figures in earlier weekly progress reports (WPR-6 through WPR-10) that were superseded or corrected during the final data reconciliation audit (04/10/2026 – 14/10/2026):

Earlier Report / Artifact	Previously Reported Claim	Reconciled Ground Truth & Reason for Revision
WPR-6 (Results), WPR-7 (Discussion), WPR-8 (Conclusion)	Sample counts of 755 (WPR-6), 1,359 (WPR-7), 1,245 (WPR-8), and 1,855 (interim SQLite); UP mean CRS 14.51 > Bihar 9.34; UP avg fluoride 1.73 mg/L and nitrate 49.82 mg/L.	Superseded by the frozen 1,741-sample canonical corpus (966 UP, 775 Bihar; 6,792 readings) from data/raw/wqmis/records.jsonl (1,822 crawled records minus 81 empty payloads). The 1,855/1,853 SQLite counts included 114 synthetic/CGWB prototype rows that were purged.
WPR-7, WPR-8, WPR-9 (v1)	Khokha village (Purnia, ID 7568) cited as a live 'chronic hotspot' with watchlist score 12.5.	Withdrawn as a live sample verdict. Khokha has 166 iron events in the 2009–2012 registry (highest in the registry) but 0 samples in the 2022 corpus; village names exist in only 21/1,741 samples (1.21%).
WPR-8, WPR-9 (v1)	Peak fluoride reading of 12.60 mg/L attributed to Agra district.	Corrected to 'Not available' district. Raw record U715825L405S1002693 in records.jsonl has state 'Uttar Pradesh' and district 'Not available' (one of 354 unassigned-district samples). Agra has 43 samples (mean CRS 60.13, max fluoride 1.58 mg/L in U1151455L303S1013828; 3.29 mg/L is in U715825L405S1005141, also district Not available).
WPR-9 (v1) pre-repair	467 sulphate readings mis-tagged as pH during initial ingestion, inflating Bihar pH exceedances.	Fixed in backend/etl/load_wqmis.py (defect B1) and repair_readings.py. True pH exceedances outside 6.5–8.5 are 0/467 in Bihar and 2/902 (0.2%) in UP.
WPR-10 (Submitted 30/09) & early WPR-11 draft	Cited a '47,243-sample production scalability dataset' across 199 districts (3,348 Critical, 8,353 High, 34,637 Low) and a Svelte 5 frontend.	Retracted and purged on 04/10/2026 (docs/evidence/RECONCILIATION.md). An audit found 45,502 bulk-inserted rows had no traceable source file in data/raw/wqmis/records.jsonl; the table was truncated and reloaded strictly from the 1,822 verifiable records in records.jsonl (1,741 valid samples). The Svelte prototype was replaced by the React dashboard (dashboard/src/App.jsx, commit adf5f0f).

Appendix B: Verified Claims Ledger

Claim / Metric in Report	Verified Value	Exact Source (Script, SQL Query, or File Line)	Audit Status
Raw crawled WQMIS reports	1,822 lines (1,741 valid, 81 empty)	wc -l data/raw/wqmis/records.jsonl	VERIFIED
Canonical scored samples	1,741 (UP: 966, Bihar: 775)	SELECT COUNT(*) FROM water_samples GROUP BY state;	VERIFIED
Discrete parameter readings	6,792 (8 params: 6,318; SO4: 467; TC: 7)	SELECT COUNT(*) FROM readings; / records.jsonl	VERIFIED
Historical registry (2009–2012)	100,690 distinct events, 14,006 villages	SELECT COUNT(*) FROM historical_contamination;	VERIFIED
Model v1 Bands (Fe = 0.3 mg/L)	89 Crit, 392 High, 28 Med, 1,232 Low	SELECT band, COUNT(*) FROM risk_scores GROUP BY band;	VERIFIED
Model v1 State Mean / Median CRS	Bihar: 18.36 / 0.00 | UP: 12.68 / 4.08	backend/app/api/meta.py / docs/api_meta.json	VERIFIED
Sensitivity Bands (Fe = 1.0 mg/L)	5 Crit, 139 High, 30 Med, 1,567 Low	backend/app/api/meta.py (sensitivity.iron_1_0)	VERIFIED
Model v2 Bands (TOTAL_BIS_WEIGHT=6.0)	3 Crit, 54 High, 125 Med (7.2%), 1,559 Low	backend/app/engine/scoring.py (commit 7331a9b)	VERIFIED
Madhepura iron-only cohort	70 samples (0.318–0.478 mg/L, med 0.382)	records.jsonl (district='Madhepura', n_params=1)	VERIFIED
Agra fluoride cluster	43 samples (mean 60.13; 31 fluoride worst)	records.jsonl (district='Agra')	VERIFIED
Peak fluoride reading	12.60 mg/L (U715825L405S1002693)	records.jsonl (state='UP', district='Not available')	VERIFIED
Unassigned district samples	354 (20.3%: 339 UP, 15 Bihar)	docs/api_meta.json (unassigned_district_samples)	VERIFIED
Village name coverage	21 of 1,741 samples (1.21%)	docs/api_meta.json (village_name_coverage_pct)	VERIFIED
Arsenic readings & zeros	259 tested (0 >0.01, max 0.009, 150 zeros)	records.jsonl (L794=69, L234=27, L269=19, L268=13)	VERIFIED
Automated pytest suite	29 tests (28 passed, 1 skipped offline; 29 w/ DB)	backend/tests/ (6 test files, 29 def test_ functions)	VERIFIED